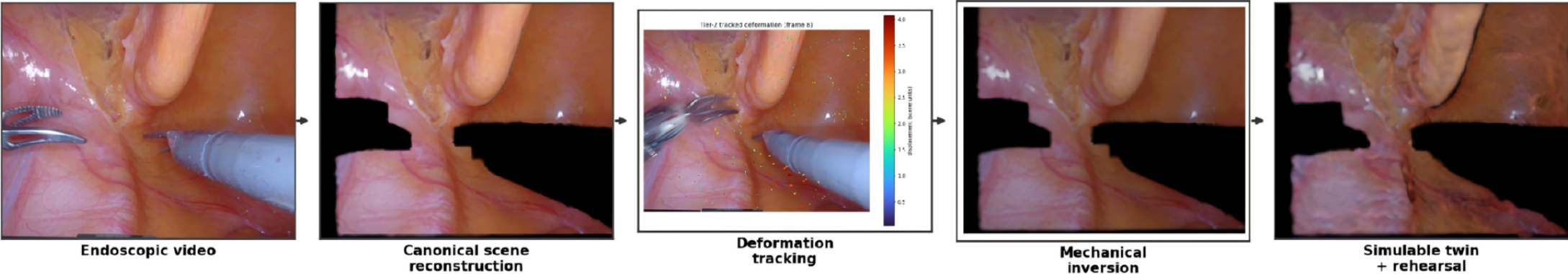
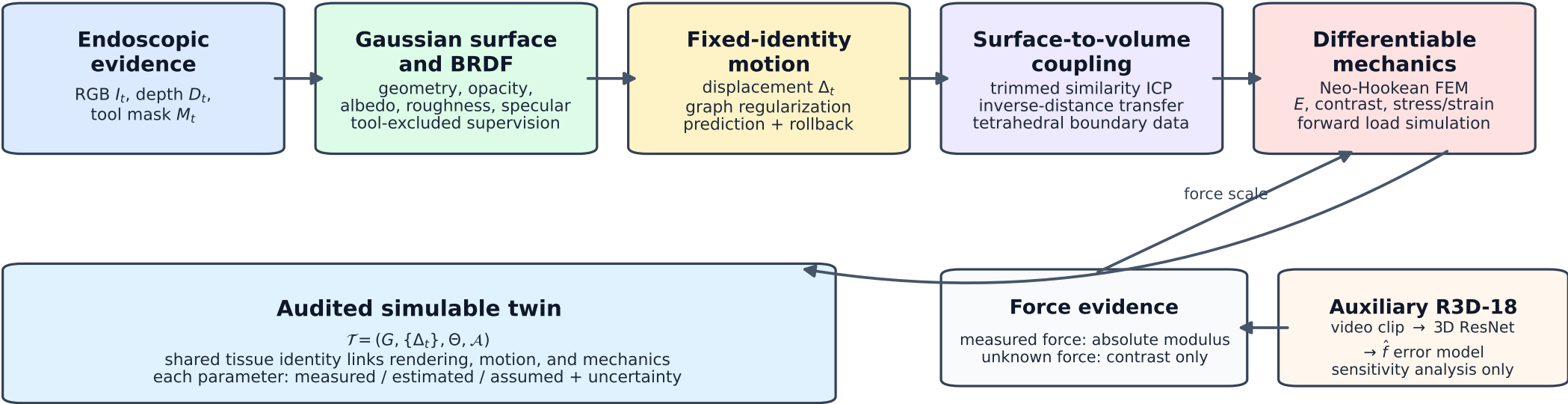


(a) Observable sequence and twin outputs



(b) Co-registered computational graph and force-dependent identifiability



Core contribution: an explicit observation-to-simulation interface; the renderer, R3D-18 backbone, and constitutive law are established components.